

MAT 324 HW05

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Problem 1. Let $(X, \mathcal{S}, \mu)$ be a measure space and $f: X \rightarrow [-\infty, \infty]$ be μ -integrable. Show that $f \geq 0$ almost everywhere on X if and only if $\int_A f \, d\mu \geq 0$ for all $A \in \mathcal{S}$.

Solution. If $f \geq 0$ almost everywhere on X , let $A \in \mathcal{S}$ and let $X' = \{x \in X : f(x) \geq 0\}$, so that $\mu(A \setminus X') = 0$ and $f \geq 0$ on $A \cap X'$. Then

$$\int_A f \, d\mu = \underbrace{\int_{A \cap X'} f \, d\mu}_{\geq 0} + \underbrace{\int_{A \setminus X'} f \, d\mu}_{=0} \geq 0.$$

Conversely suppose that f is negative on a set $A \in \mathcal{S}$ of positive measure. By Axler's exercise 3A #3, $\int_A -f \, d\mu > 0 \iff \int_A f < 0$, which proves the *only if* direction. $\square$

Problem 2 (Axler 3B #3). Suppose λ is the Lebesgue measure on $\mathbb{R}$ and $f: \mathbb{R} \rightarrow \mathbb{R}$ is a Borel measurable function such that $\int |f| \, d\lambda < \infty$. Define $g: \mathbb{R} \rightarrow \mathbb{R}$ by $g(x) = \int_{(-\infty, x)} f \, d\lambda$. Prove that g is uniformly continuous on $\mathbb{R}$.

Solution. The integral bound tells us $\int_X f_{\pm} \leq \int_X f < \infty$, so $\int_X f$ is well-defined and finite. Let $\varepsilon > 0$. By Axler's 3.28 there exists $\delta > 0$ such that

$$g(x) - g(y) = \int_{(x, y)} f \, d\lambda < \varepsilon$$

for all $x, y \in \mathbb{R}$ such that $x \leq y$ and $\lambda((x, y)) = y - x < \delta$, so g is uniformly continuous. $\square$

Problem 3 (Axler 3B #4). (a) Suppose $(X, \mathcal{S}, \mu)$ is a measure space with $\mu(X) < \infty$. Suppose that $f: X \rightarrow [0, \infty)$ is a bounded $\mathcal{S}$ -measurable function. Prove that

$$\int f \, d\mu = \inf \left\{ \sum_{j=1}^m \mu(A_j) \sup_{A_j} f : A_1, \dots, A_m \text{ is an } \mathcal{S}\text{-partition of } X \right\}.$$

(b) Show that the conclusion of (a) can fail if the hypothesis that f is bounded is replaced by the hypothesis that $\int f \, d\mu < \infty$.

(c) Show that the conclusion of (a) can fail if the condition that $\mu(X) < \infty$ is deleted.

Solution. We refer to the infimum of the upper Lebesgue sums as the **upper Lebesgue integral**, in contrast with the standard **(lower) Lebesgue integral** that we defined with lower sums.

(a) Since f is bounded, there exists $C > 0$ such that $f(x) < C$ for all x . Splitting f as $C - (C - f)$ yields

$$\begin{aligned} \int_X f &= C\mu(X) - \int_X (C - f) = C\mu(X) - \sup \left\{ \sum_{j=1}^m \mu(A_j) \inf_{A_j} (C - f) : A_1, \dots, A_m \text{ is an } \mathcal{S}\text{-partition of } X \right\} \\ &= C\mu(X) - \sup \left\{ C\mu(X) - \sum_{j=1}^m \mu(A_j) \sup_{A_j} f : A_1, \dots, A_m \text{ is an } \mathcal{S}\text{-partition of } X \right\} \\ &= \cancel{C\mu(X)} - \cancel{C\mu(X)} + \inf \left\{ \sum_{j=1}^m \mu(A_j) \sup_{A_j} f : A_1, \dots, A_m \text{ is an } \mathcal{S}\text{-partition of } X \right\}. \end{aligned}$$

(b) Consider $(0, 1]$ with the Lebesgue measure λ and let $f: (0, 1] \rightarrow \mathbb{R}$ be the Lebesgue measurable function $f(x) = 1/\sqrt{x}$. Let P be a Lebesgue partition of $(0, 1]$ (in analogy with “ $\mathcal{S}$ -partition”) and let $P' = \{I \in P : 0 \in \bar{I}\}$; P' is nonempty because $0 \in \bigcup_{I \in P} \bar{I} = \bigcup_{I \in P'} \bar{I}$, with the equality because P is finite. The union $A = \bigcup_{I \in P'} I$ has $\lambda(A) > 0$ because either $A = (0, 1]$ or we can find $I \in P \setminus P'$ with $\inf I \neq 0$ minimal, so $(0, \inf I) \subseteq A$; thus some $J \in P'$ has $\lambda(J) > 0$. Since $\lim_{x \rightarrow 0} f(x) = \infty$, $\sup_J f = \infty$, so

$$\sum_{I \in P} \lambda(I) \sup_I f \geq \lambda(J) \sup_J f = \infty \implies \inf \left\{ \sum_{I \in P} \lambda(I) \sup_I f : P \text{ is a Lebesgue partition of } X \right\} = \infty.$$

On the other hand, f is the limit of the increasing sequence of functions $f_n = f \cdot \mathbf{1}_{[1/n, 1]}$ that are Riemann integrable on $[0, 1]$, so by monotone convergence and Axler’s 3.34,

$$\int_{(0, 1]} f \, d\lambda = \lim_{n \rightarrow \infty} \left(\int_{[1/n, 1]} f_n \, d\lambda \right) = \lim_{n \rightarrow \infty} \left(\int_{1/n}^1 \frac{1}{\sqrt{x}} \, dx \right) = \lim_{n \rightarrow \infty} \left(\frac{2}{3} - \frac{2}{3} \cdot \left(\frac{1}{n} \right)^{3/2} \right) = \frac{2}{3} \neq \infty.$$

(c) Consider $[1, \infty)$ endowed with the Lebesgue measure λ and let $g: [1, \infty) \rightarrow [0, \infty)$ be the bounded Lebesgue measurable function $g(x) = 1/x^2$. Let P be a Lebesgue partition of $[0, \infty)$. Since the union of the sets in P has infinite Lebesgue measure, some $J \in P$ has $\lambda(J) = \infty$. Also, $\sup_J g > 0$ because $g > 0$, so

$$\sum_{I \in P} \lambda(I) \sup_I g \geq \lambda(J) \sup_J g = \infty \implies \inf \left\{ \sum_{I \in P} \lambda(I) \sup_I g : P \text{ is a Lebesgue partition of } X \right\} = \infty.$$

On the other hand, g is the limit of the increasing sequence of functions $g_n = g \cdot \mathbf{1}_{[1, n]}$ that are Riemann integrable on $[1, n]$, so by monotone convergence and Axler’s 3.34,

$$\int_{[1, \infty)} g \, d\lambda = \lim_{n \rightarrow \infty} \left(\int_{[1, n]} g_n \, d\lambda \right) = \lim_{n \rightarrow \infty} \left(\int_1^n \frac{1}{x^2} \, dx \right) = \lim_{n \rightarrow \infty} (1 - 1/n) = 1 \neq \infty. \quad \square$$

Problem 4. Let $(X, \mathcal{S}, \mu)$ be a measure space and $g_1, g_2, \dots : X \rightarrow [0, \infty]$ be a sequence of $\mathcal{S}$ -measurable functions.

(a) Show that $\int_X (\liminf_{n \rightarrow \infty} g_n) d\mu \leq \liminf_{n \rightarrow \infty} (\int_X g_n d\mu)$.

(b) Suppose also that the sequence $g_1, g_2, \dots$ converges (pointwise) to a function $g : X \rightarrow [0, \infty]$ and

$$\lim_{n \rightarrow \infty} \left(\int_X g_n d\mu \right) = \int_X g d\mu < \infty.$$

Show that $\lim_{n \rightarrow \infty} (\int_E g_n d\mu) = \int_E g d\mu$ for all $E \in \mathcal{S}$.

(c) Suppose in addition (to all of the above) that $f_1, f_2, \dots : X \rightarrow [-\infty, \infty]$ is a sequence of $\mathcal{S}$ -measurable functions converging to a function $f : X \rightarrow [-\infty, \infty]$ such that $|f_n| \leq g_n$ for all $n \in \mathbb{N}$. Show that

$$\int_X f d\mu = \lim_{n \rightarrow \infty} \left(\int_X f_n d\mu \right).$$

Solution. (a) For any $N \in \mathbb{N}$ we have that

$$\int_X (\inf_{n \geq N} g_n) d\mu \leq \int_X g_n d\mu \text{ for all } n \geq N \implies \int_X (\inf_{n \geq N} g_n) d\mu \leq \inf_{n \geq N} \int_X g_n d\mu$$

Since $\inf_{n \geq N} g_n$ is increasing in N , we can take supremums and use monotone convergence to get

$$\int_X \liminf_{n \rightarrow \infty} g_n = \sup_{N \rightarrow \infty} \left(\int_X (\inf_{n \geq N} g_n) d\mu \right) \leq \sup_{N \rightarrow \infty} \inf_{n \geq N} \left(\int_X g_n d\mu \right) = \liminf_{n \rightarrow \infty} \left(\int_X g_n d\mu \right).$$

(b) It suffices to show that $\liminf_{n \rightarrow \infty} (\int_E g_n d\mu) = \limsup_{n \rightarrow \infty} (\int_E g_n d\mu) = \int_E g d\mu$; to do this, we can use Fatou's lemma on the sequences $\int_{X \setminus E} g_n d\mu$ and $\int_E g_n d\mu$ in the following form:

$$\begin{aligned} \liminf_{n \rightarrow \infty} \left(\int_E g_n d\mu \right) &\leq \limsup_{n \rightarrow \infty} \left(\int_E g_n d\mu \right) = \limsup_{n \rightarrow \infty} \left(\int_X g d\mu - \int_{X \setminus E} g_n d\mu \right) \\ &= \int_X g d\mu - \liminf_{n \rightarrow \infty} \left(\int_{X \setminus E} g_n d\mu \right) \\ &\leq \int_X g d\mu - \int_{X \setminus E} g d\mu = \int_E g d\mu \leq \liminf_{n \rightarrow \infty} \left(\int_E g d\mu \right). \end{aligned}$$

(c) Analogous to Axler's 3.32, for each $E \in \mathcal{S}$ and $n \in \mathbb{N}$ we have the bound

$$\left| \int_X f_n - \int_X f \right| \leq \left| \int_{X \setminus E} f_n \right| + \left| \int_{X \setminus E} f \right| + \left| \int_E f_n - \int_E f \right| \leq \int_{X \setminus E} g_n + \int_{X \setminus E} g + \left| \int_E f_n - \int_E f \right|. \quad (*)$$

(for conciseness, we drop $d\mu$ from all the integrals here).

Case 1. Suppose that $\mu(X) < \infty$ and let $\varepsilon > 0$; then, let $\delta > 0$ be as in Axler's proof of 3.31, but with the bound on $\int_B g$ replaced by $\varepsilon/6$. By (b), if n is sufficiently large, $|\int_B g_n - \int_B g| < \varepsilon/6$, which implies $\int_B g_n < \varepsilon/3$ for all $B \in \mathcal{S}$ with $\mu(B) < \delta$. Now let $E \in \mathcal{S}$ as in Axler's proof and use $(*)$ to get

$$\left| \int_X f_n - \int_X f \right| \leq \frac{\varepsilon}{2} + \left| \int_E (f_n - f) \right| \quad (\dagger)$$

from Axler's proof if n is large enough; thus $\int_X f_n$ converges to $\int_X f$ with the same logic as Axler's proof.

Case 2. If $\mu(X) = \infty$, pick $E \in \mathcal{S}$ as in Axler's proof, but with $\varepsilon/6$ instead of $\varepsilon/4$. Again by (b), if n is sufficiently large, then $\int_{X \setminus E} g_n d\mu < \varepsilon/3$, so applying $(*)$ gives the bound $(\dagger)$, which again implies convergence of the integrals. $\square$

Problem 5. Let λ denote the Lebesgue measure on $[0, \infty)$. For each $n \in \mathbb{N}$, define

$$f_n, g_n: [0, \infty) \rightarrow \mathbb{R}, \quad f_n(x) = \frac{n^2 x e^{-n^2 x^2}}{1+x}, \quad g_n(x) = \frac{x e^{-x^2}}{1+x/n}.$$

The integrals in (a) and (b) below are improper Riemann integrals.

(a) Find $\int_0^\infty (\lim_{n \rightarrow \infty} f_n) dx$ and $\int_0^\infty (\lim_{n \rightarrow \infty} g_n) dx$.

(b) Show that

$$\lim_{n \rightarrow \infty} \left(\int_0^\infty f_n dx \right) = \lim_{n \rightarrow \infty} \left(\int_0^\infty g_n dx \right)$$

and find this limit.

(c) Show that there exists no Lebesgue measurable function $F: [0, \infty) \rightarrow [0, \infty]$ such that

$$f_n \leq F \text{ a.e. for all } n \in \mathbb{N} \text{ and } \int_{[0, \infty)} F d\lambda < \infty.$$

Solution. (a) We can compute the limits (using l'Hôpital's rule for f)

$$\begin{aligned} f &:= \lim_{n \rightarrow \infty} f_n = \frac{1}{1+x} \cdot \lim_{n \rightarrow \infty} \frac{n^2 x}{e^{n^2 x^2}} = \frac{1}{1+x} \cdot \lim_{n \rightarrow \infty} \frac{\cancel{n^2} x}{\cancel{n^2} \cdot 2x e^{n^2 x^2}} = 0, \\ g &:= \lim_{n \rightarrow \infty} g_n = \frac{x e^{-x^2}}{1 + \lim_{n \rightarrow \infty} x/n} = x e^{-x^2}. \end{aligned}$$

Then, the integrals of f and g are

$$\int_0^\infty f dx = \boxed{0}, \quad \int_0^\infty g dx = \lim_{c \rightarrow \infty} \left(\int_0^c g dx \right) = \lim_{c \rightarrow \infty} \left(-\frac{1}{2} e^{-c^2} + \frac{1}{2} e^0 \right) = \boxed{\frac{1}{2}}.$$

(b) For $n \in \mathbb{N}$, let $u = nx$. Then

$$\int_0^\infty g_n dx = \int_0^\infty \frac{u e^{-u^2}}{1+u/n} du = \int_0^\infty \frac{n x e^{-n^2 x^2}}{1+x} \cdot n dx = \int_0^\infty f_n dx,$$

so $\int_0^\infty f_n dx$ and $\int_0^\infty g_n dx$ have the same limit.

For any Lebesgue measurable function $h: [0, \infty) \rightarrow \mathbb{R}$ such that the Lebesgue integral $\int_{[0, \infty)} h d\lambda$ and improper Riemann integral $\int_0^\infty h dx$ are well-defined, the two integrals must be equal; to show this, we can pass the Riemann integral to the Lebesgue integral by Axler's 3.34 and use monotone convergence in the following fashion:

$$\int_0^\infty h dx = \lim_{c \rightarrow \infty} \left(\int_0^c h dx \right) = \lim_{c \rightarrow \infty} \left(\int_{[0, c]} h d\lambda \right) = \lim_{c \rightarrow \infty} \left(\int_{[0, \infty)} h \cdot \mathbf{1}_{[0, c]} d\lambda \right) = \int_{[0, \infty)} h d\lambda \quad (*)$$

To finish, g_n is an increasing sequence (because $1+x/n$ is decreasing in n), so by monotone convergence,

$$\lim_{n \rightarrow \infty} \left(\int_0^\infty f_n dx \right) = \lim_{n \rightarrow \infty} \left(\int_0^\infty g_n dx \right) \stackrel{(*)}{=} \lim_{n \rightarrow \infty} \left(\int_{[0, \infty)} g_n d\lambda \right) \stackrel{\text{MCT}}{=} \int_{[0, \infty)} g d\lambda \stackrel{(*)}{=} \int_0^\infty g dx \stackrel{(a)}{=} \boxed{\frac{1}{2}}.$$

(c) Suppose there exists such a function $F: [0, \infty) \rightarrow [0, \infty]$. Let $\tilde{F}(x) = F$ at all $x \in X$ such that $F(x) \geq f_n(x)$ and $\tilde{F}(x) = \infty$ for all other x . Since F and $\tilde{F}$ only disagree on a set of measure zero by hypothesis, they have the same (finite) integral, and $\tilde{F} \geq f_n = |f_n|$ for all n ; thus by the dominated convergence theorem (where $\tilde{F}$ dominates the sequence f_n),

$$\frac{1}{2} \stackrel{(b)}{=} \lim_{n \rightarrow \infty} \left(\int_{[0, \infty)} f_n d\lambda \right) = \int_{[0, \infty)} f d\lambda \stackrel{(*)}{=} \int_0^\infty f dx \stackrel{(a)}{=} 0.$$

This is a contradiction, so F cannot exist. □

Problem 6. Show that the function

$$f: [0, \infty) \rightarrow \mathbb{R}, \quad f(x) = \begin{cases} \frac{\sin x}{x} & x > 0 \\ 1 & x = 0 \end{cases}$$

has an improper Riemann integral over $[0, \infty)$, but is not Lebesgue integrable on $[0, \infty)$.

Solution. Let $I_n = [(n-1)\pi, n\pi]$ and

$$a_n = \int_{I_n} f \, d\lambda = \int_{(n-1)\pi}^{n\pi} f \, dx,$$

noting that f is continuous hence Riemann integrable on each I_n . For $n > 1$, we can bound

$$\frac{1}{n\pi} = \left| \int_{(n-1)\pi}^{n\pi} \frac{\sin x}{n\pi} \, dx \right| \leq \underbrace{\left| \int_{(n-1)\pi}^{n\pi} \frac{\sin x}{x} \, dx \right|}_{|a_n|} \leq \left| \int_{(n-1)\pi}^{n\pi} \frac{\sin x}{(n-1)\pi} \, dx \right| = \frac{1}{(n-1)\pi} \quad (**)$$

with the lower bound holding for $n = 1$ as well. Since f is nonnegative exactly when $\sin x \geq 0$, equivalently $x \in I_{2n-1}$ for some $n \in \mathbb{N}$ (and nonpositive if and only if $x \in I_{2n}$ for some $n \in \mathbb{N}$),

$$f_+ = \sum_{n=1}^{\infty} f \cdot \mathbf{1}_{I_{2n-1}}, \quad f_- = \sum_{n=1}^{\infty} f \cdot \mathbf{1}_{I_{2n}}.$$

In particular, by HW04 #5a,

$$\int_{[0, \infty)} f_+ \, d\lambda = \sum_{n=0}^{\infty} a_{2n+1} \geq \sum_{n=0}^{\infty} \frac{1}{(2n+1)\pi} = \infty, \quad \int_{[0, \infty)} f_- \, d\lambda = \sum_{n=1}^{\infty} a_{2n} \geq \sum_{n=1}^{\infty} \frac{1}{2n\pi} = \infty.$$

Thus $\int f_+ \, d\lambda$ and $\int f_- \, d\lambda$ are both ∞ , so f is not Lebesgue integrable on $[0, \infty)$.

For Riemann integrability, define $F: [0, \infty) \rightarrow \mathbb{R}$ by

$$F(y) = \int_0^y f \, dx.$$

Given $c \in [0, \infty)$, let n be the largest integer such that $c \in I_n$ (so if $k \in \mathbb{N}$ such that $c = k\pi$ then $n = k+1$). Since f is either always nonpositive or always nonnegative on I_n , F is monotonic on I_n , meaning $F(c)$ lies between $F(n)$ and $F(n+1)$. Thus

$$\lim_{c \rightarrow \infty} F(c) = \lim_{n \rightarrow \infty} F(n) = \sum_{k=1}^{\infty} a_k = \sum_{k=1}^{\infty} (-1)^{k+1} |a_k|.$$

Since $|a_k| \leq \frac{1}{(k-1)\pi} \leq |a_{k-1}|$ by $(**)$ for all $k \in \mathbb{N}$, the sequence $|a_k|$ converges to 0, so the sum on the right-hand side converges by the alternating series test. This means

$$\lim_{c \rightarrow \infty} F(c) = \int_0^{\infty} f \, dx$$

is finite, hence f has an improper Riemann integral on $[0, \infty)$. □